INTERNSHIP PROJECT REPORT

Train Schedule Analysis and Interactive Route Enquiry System Using Python

Sysslan IT Solutions
Data Analysis Internship Project

Prepared by	Chunduri Manikanta
Department	AI & Robotics
Project dataset	Dataset1.csv
Implementation	Python / Google Colab

Prepared from the submitted Google Colab notebook and generated analysis outputs.

Abstract

This project analyses a railway timetable dataset using Python and develops an interactive, dataset-based route enquiry system. The work progresses through six levels: basic dataset review, schedule processing, data-quality checks, exploratory analysis, advanced pivot and cross-tabulation analysis, and interactive source-to-destination train enquiry. The dataset contains 186,074 schedule records, 11,113 distinct train numbers and 8,147 station codes. The workflow applies explicit rules for station sequence, missing and placeholder times, midnight rollover, repeated station visits and questionable schedule records. The enquiry system searches direct trains in the correct station order, reports estimated segment duration, displays intermediate stations and supports pagination and CSV export. It is a static timetable tool, not a live train status service.

1. Introduction

Railway timetable data contains train identifiers, station sequences, arrival and departure times and route distances. Analysing these fields can reveal route patterns, station service frequency, journey-duration trends and data-quality issues. This project follows a staged workflow so the final enquiry feature is based on inspected, processed and validated records.

2. Objectives

- Understand dataset structure and characteristics.
- Identify origin, terminus and recorded station counts for each train.
- Standardise schedule times and calculate journey durations with midnight rollover.
- Classify routes using data-derived distance thresholds and calculate station frequency.
- Detect missing values, duplicates, sequence issues and time regressions.
- Perform exploratory and advanced analysis using tables and charts.
- Build an interactive direct-train enquiry system.

3. Dataset Description

Input file: Dataset1.csv. The 12 original columns are SN, Train_No, Station_Code, 1A, 2A, 3A, SL, Station_Name, Route_Number, Arrival_time, Departure_Time and Distance. The notebook reports 186,074 rows, 11,113 distinct trains and 8,147 station codes. Text-like identifiers are read as text to preserve leading zeros and station codes.

Characteristic	Value
Schedule rows	186,074
Original attributes	12
Distinct trains	11,113
Distinct station codes	8,147
Median recorded stations/train	15
Mean recorded stations/train	16.7

4. Tools and Libraries

Python and Google Colab are used for implementation. Pandas and NumPy support data processing and calculations; Matplotlib and Seaborn support visualization; IPython display and Colab utilities support notebook interaction and file upload/download; Python standard-library modules support file handling, time operations, hashing, ZIP packaging and validation.

5. Methodology and Implementation

5.1 Level 1 – Basic Data Review

The notebook retains the original data separately, orders records within each train by SN (using original file position to break ties), and derives Stop_Pos and N_Stations. The first and last records identify origin and terminus. Intermediate stops equal max(total recorded stations − 2, 0). Repeated visits remain separate records. Train 53041 is listed from HOWRAH JN. (HWH) to JAYNAGAR (JYG) with 118 recorded stations.

5.2 Level 2 – Simple Data Processing

Arrival and departure times are validated as HH:MM:SS and classified as valid, missing or invalid. Origin arrival and terminus departure placeholders are treated as not applicable. Ambiguous meaningful endpoint 00:00:00 values are treated as unknown. Times are unrolled onto a continuous timeline; a backward shift greater than 12 hours indicates a day rollover, while smaller backward movements are flagged as regressions. Journey duration is first-station departure to final-station arrival.

Route distance is final Distance minus initial Distance. Its unit is not confirmed. Short/Medium/Long boundaries are data-derived terciles: Short up to 49 units, Medium up to 171 units, Long above 171 units. Route_Number is constant and not used. Station frequency is calculated using both distinct trains and total record occurrences.

5.3 Level 3 – Data Quality Checks

Exact duplicate rows are removed and reported. Records that only appear duplicated after normalisation and repeated station visits are flagged and retained. Critical issues include duplicate or invalid SN and trains with fewer than two records; warnings include SN gaps, distance issues and time regressions; informational findings include loops and other non-critical observations.

Check	Affected trains
Duplicate SN	0
Blank/non-numeric SN	0
Fewer than 2 records	0
SN gaps	5
Decreasing distance	0
Missing distance	0
Time regressions	7
Repeated station codes	20

The cleaned dataset retains the working records and adds standardised fields and quality flags. The notebook verifies that the original upload remains unchanged.

5.4 Level 4 – Basic Analysis and Visualization

The analysis compares reliable journey durations by route category, ranks high-traffic stations and examines distributions of journey duration, train size and station frequency. Reliable durations exist for 11,102 classified trains; mean duration is 7.2 hours, median 2.2 hours and maximum 82 hours 45 minutes. A total of 2,537 reliable journeys cross midnight.

Category	Trains	Mean duration	Median duration
Short	3,900	0.88 h	0.92 h
Medium	3,505	2.60 h	2.33 h
Long	3,697	18.26 h	13.67 h

5.5 Level 5 – Advanced Analysis and Visualization

Pivot tables summarize station-by-category distinct-train counts and route-category duration, distance and train-size statistics. Cross-tabulations compare the 25 busiest stations by category, category by train-size quartile, and traffic tier by category. The busiest station is CST-MUMBAI (CSMT), served by 1,027 distinct trains, followed by KALYAN JN (828), THANE (796), SEALDAH (745) and CHENNAI BEAC (738). Mean station frequency is 22.8 while the median is 10, showing a skewed service distribution. This does not alone prove a hub-and-spoke network topology.

Spearman correlation between route distance and duration is 0.98; between route distance and recorded station count it is 0.64. Distance units are unconfirmed, so these are dataset-relative measures, not kilometres or physical speed.

5.6 Level 6 – Interactive Train Route Enquiry System

The enquiry interface uses cleaned, ordered data. A direct train must call at the source and later at the destination. The destination position must be greater than the source position, preventing wrong-direction matches. Looping train visit pairings are retained. Users can search by station code, exact name or partial name/code, sort results, page through all matches, export all results to CSV and view every intermediate station. Warnings appear when relevant to the searched stretch. The tool uses static timetable information and has no live railway API.

6. Key Findings

- 186,074 schedule rows represent 11,113 trains and 8,147 station codes.
- The median train has 15 recorded station entries; the longest-listed train has 118.
- CST-MUMBAI (CSMT) is busiest with 1,027 distinct trains.
- The 20 busiest stations account for 5.7% of all station occurrences.
- 53 stations are served by one train; median station service is 10 trains.
- 11,102 reliable journey durations are available; 11 are unreliable.
- Mean duration is about 0.9 h, 2.6 h and 18.3 h for Short, Medium and Long groups.
- Twenty trains repeat at least one station code; five have SN gaps; seven have time regressions.
- 2,537 reliable journeys cross midnight.

7. Validation and Testing

The notebook includes assertion tests for time parsing, formatting, rollover, regression detection, route classification, pagination, real-data consistency and enquiry behaviour. The supplied validation_tests.csv records 35 passing tests out of 35 listed tests. Tests are evidence for implemented logic, not proof that the source timetable is complete or live.

8. Limitations

- Static timetable only; no live status, delay or platform updates.
- Distance unit is not confirmed; neutral units are used.
- Route categories are based on dataset-derived terciles, not official railway definitions.
- Endpoint midnight placeholders can be ambiguous.
- Midnight rollover is inferred using a configured time threshold; no calendar dates are available.
- Findings describe this dataset and should not be generalized to the full railway network without authoritative broader data.

9. Conclusion

The project completes the six-stage workflow specified in the internship task: basic review, processing, quality checks, basic analysis, advanced analysis and interactive enquiry. It converts raw timetable records into structured reports, quality indicators, visualizations and an interactive lookup tool. It demonstrates practical Python skills in data loading, cleaning, grouping, time processing, statistical analysis, visualization and interactive data lookup.

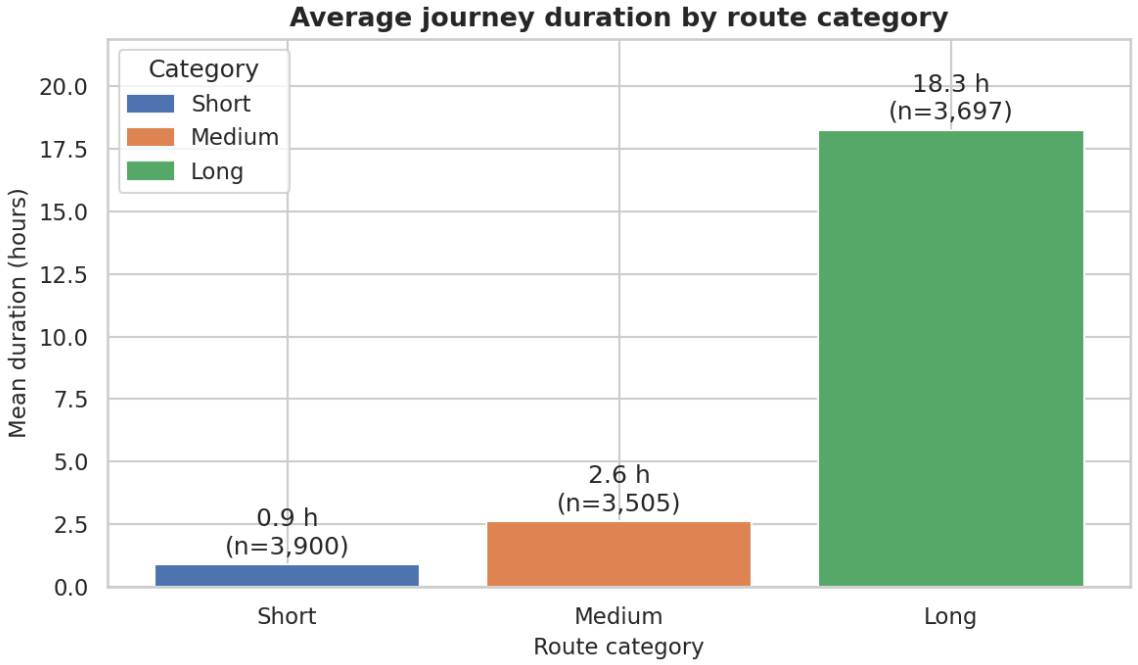
10. Deliverables

- Google Colab notebook: Train_Schedule_Analysis_(1).ipynb
- ZIP archive with 35 generated files, including CSV reports, insight summary, validation results and 12 charts.
- Cleaned train dataset and data-quality reports.
- Level 1–5 analysis tables, pivot tables, cross-tabulations and visualizations.
- Level 6 interactive route enquiry and CSV export capability.

Appendix A – Generated Charts

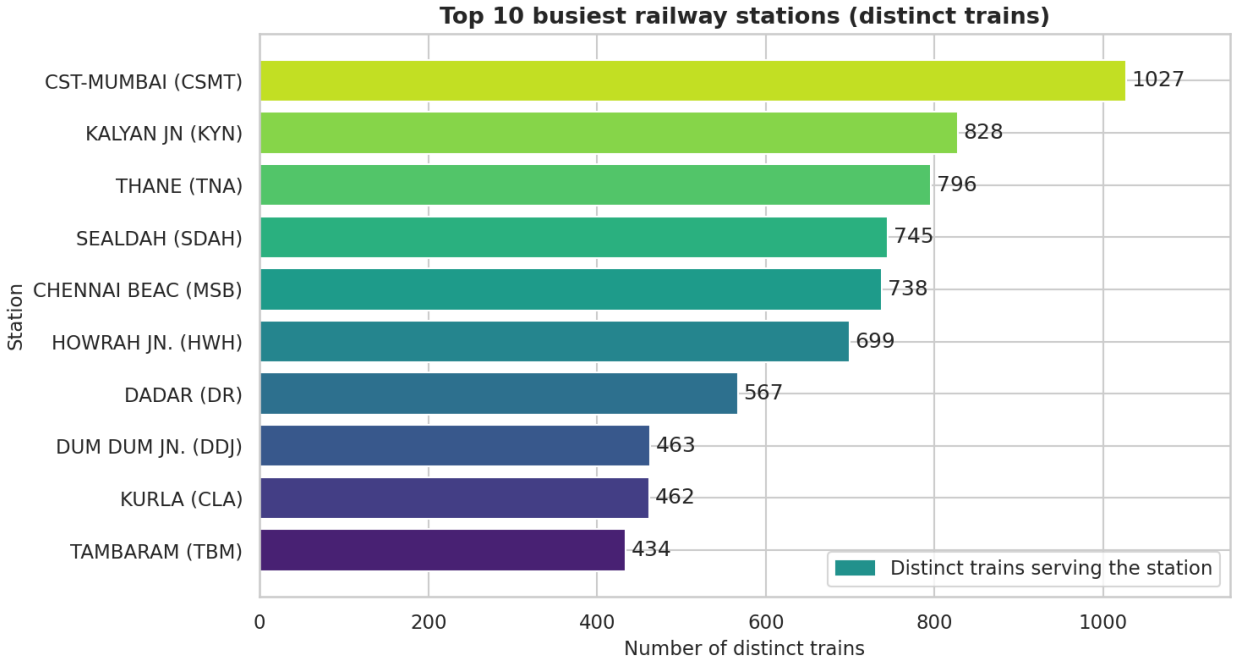
Average journey duration by route category

Compares mean duration for reliable journeys in the Short, Medium and Long groups.



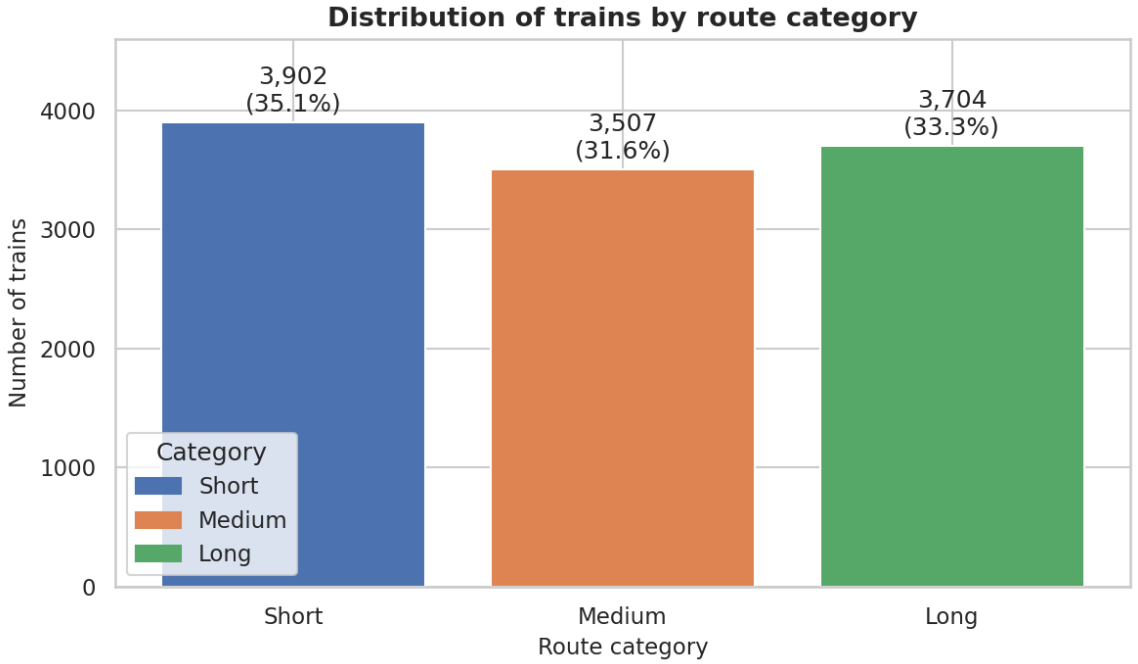
Top 10 busiest stations

Ranks stations by distinct trains calling at each station.



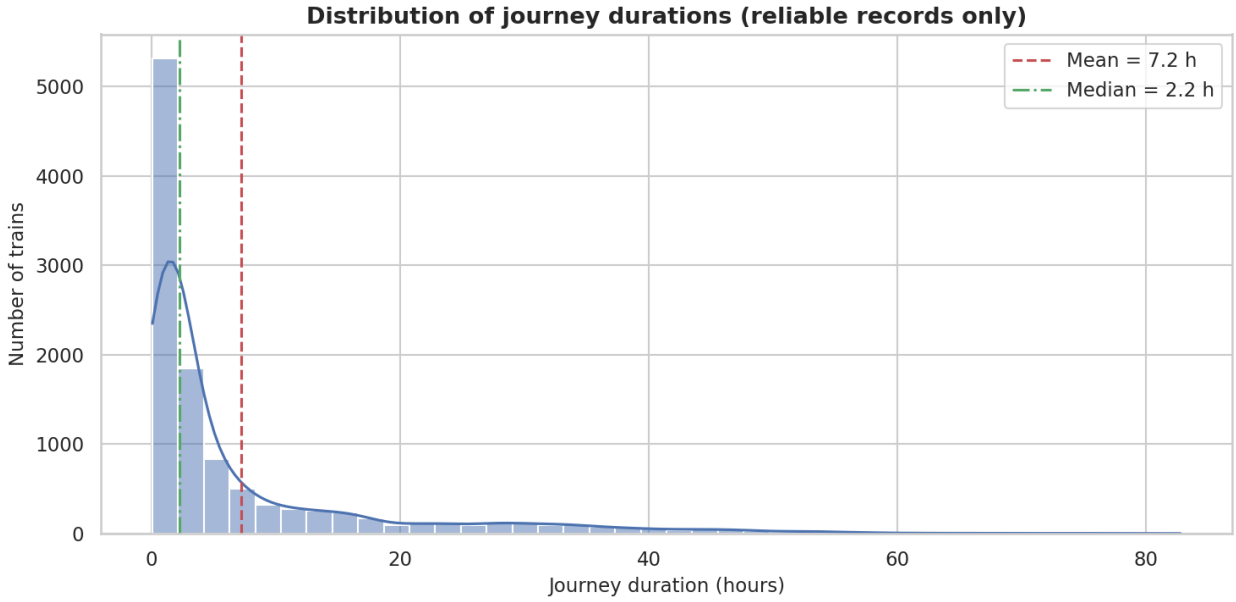
Trains by route category

Shows the number of trains assigned to each distance-based category.



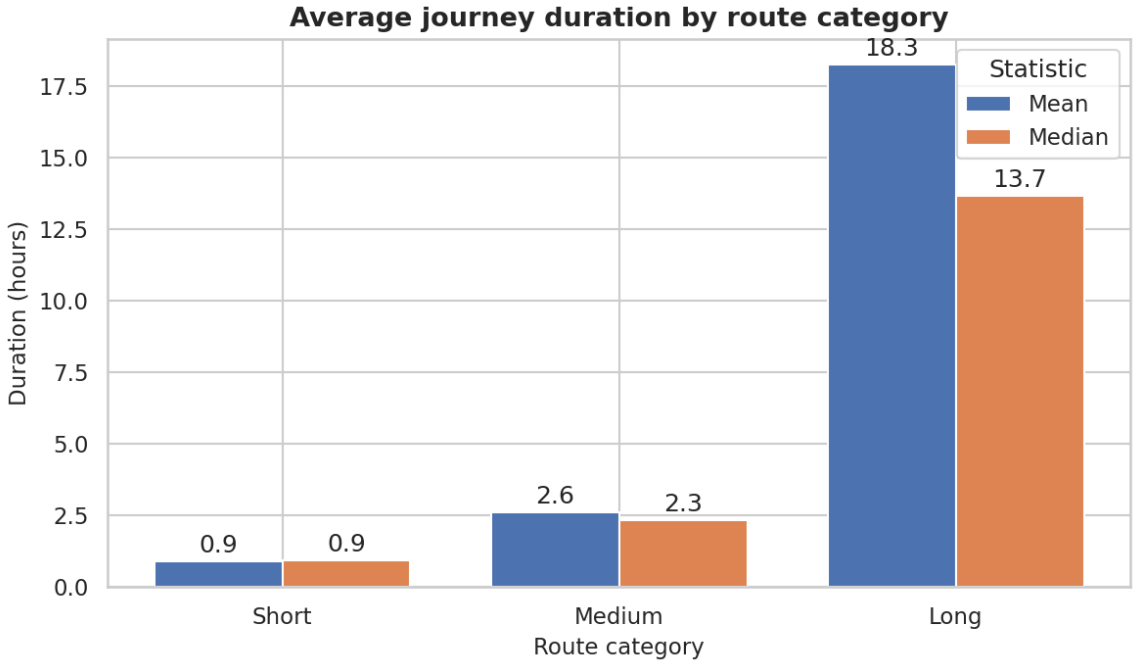
Journey duration distribution

Shows the distribution of reliable end-to-end journey durations.



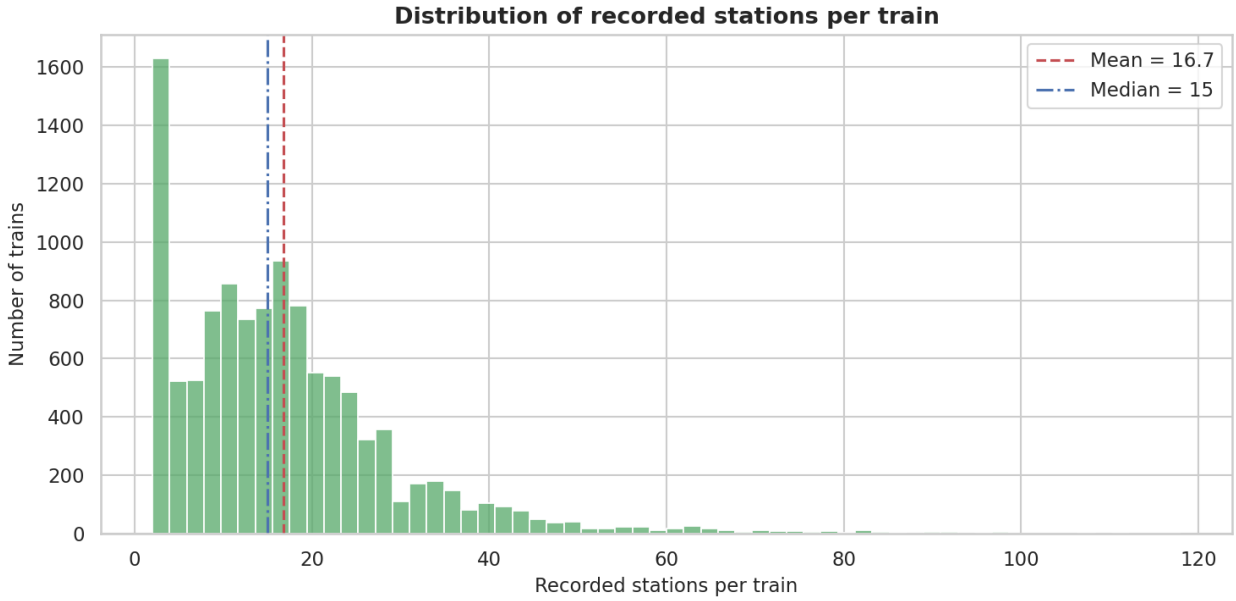
Mean and median duration by category

Compares mean and median journey duration for each category.



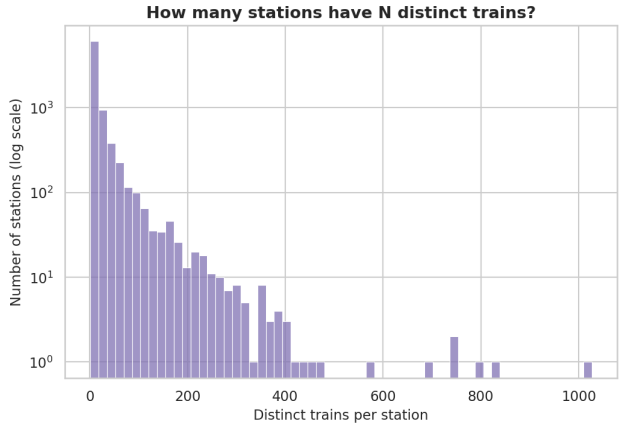
Stations recorded per train

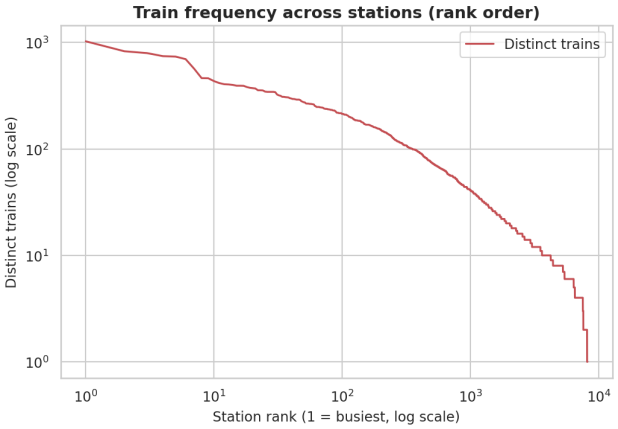
Shows how the number of recorded station entries varies across trains.



Train frequency across stations

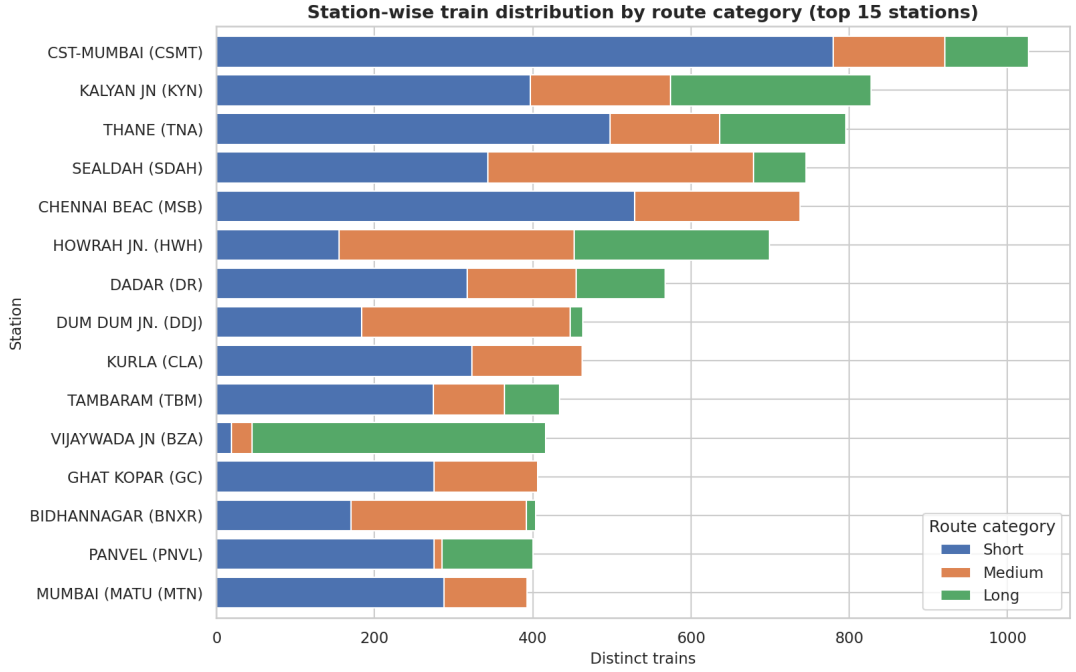
Shows the distribution of distinct train frequency by station.





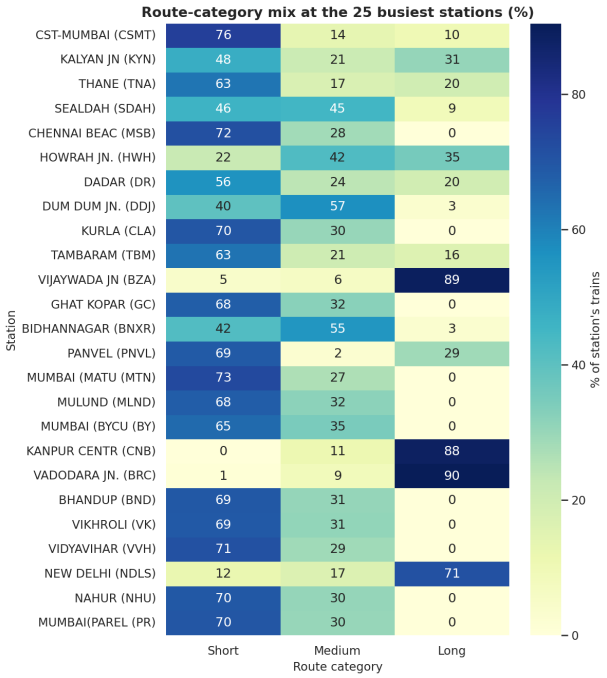
Station distribution by route category

Stacked comparison of route-category service at stations.



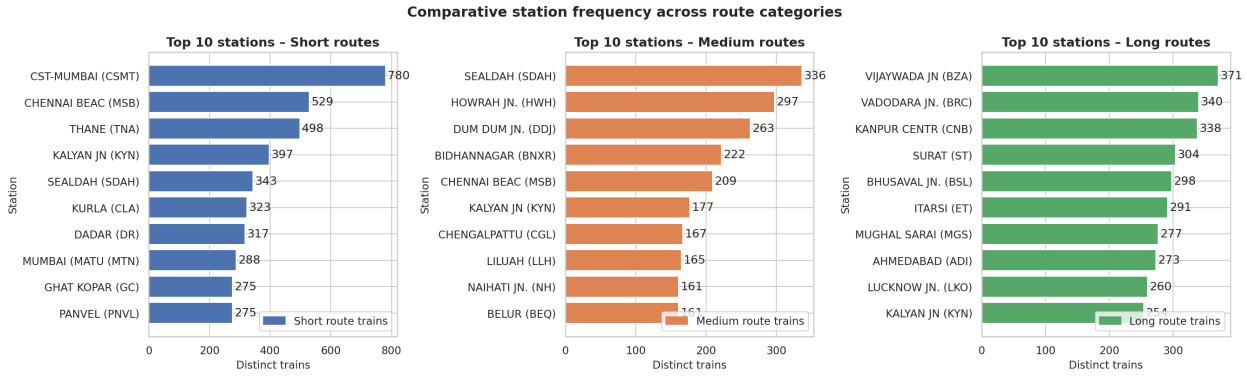
Station and route-category heat map

Visualizes the category composition across selected busy stations.



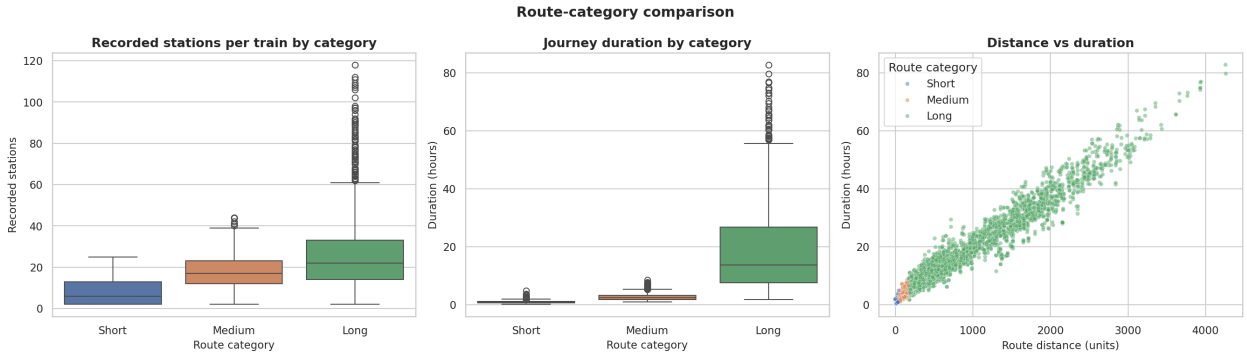
Top stations within each category

Compares the most-served stations for each route category.



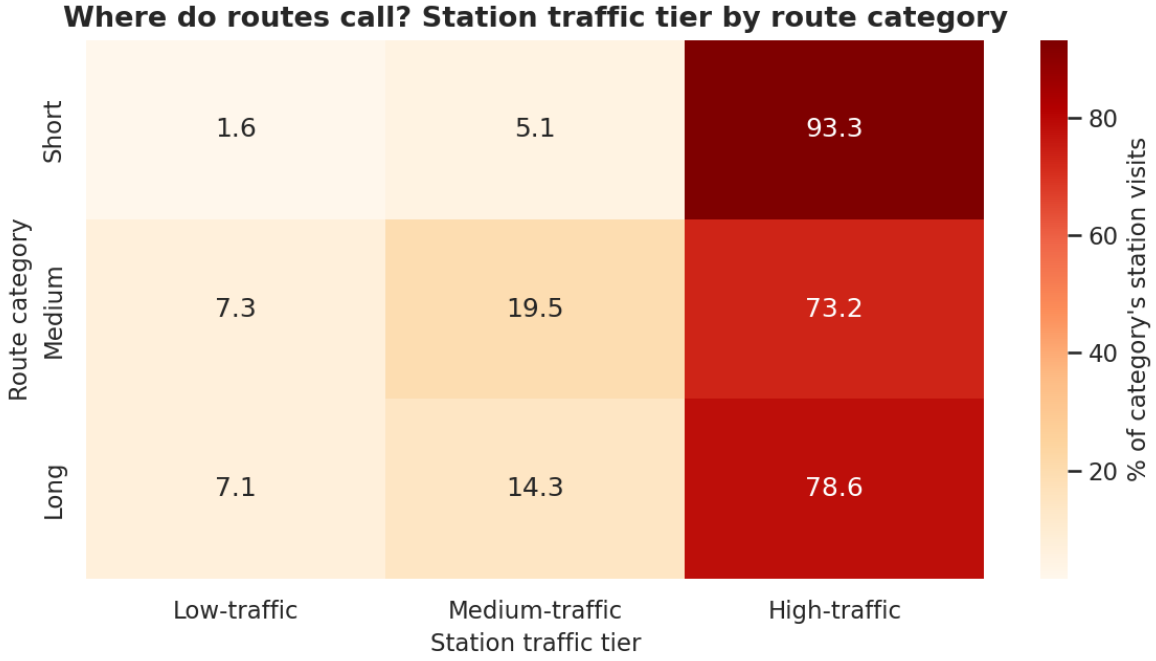
Route-category comparison

Compares train size, duration and distance-related measures by category.



Traffic-tier and route-category heat map

Shows route-category visits across low-, medium- and high-traffic station tiers.


Where do routes call? Station traffic tier by route category
Route category
Short
Medium
Long
1.6
5.1
93.3
7.3
19.5
73.2
7.1
14.3
78.6
Low-traffic
Medium-traffic
High-traffic
Station traffic tier
80
60
40
20
% of category's station visits